

Task Details

Description

This assignment relates to how modern systems operate through interconnected components, external services, APIs, messaging, and cloud environments. It requires you to draw together key learnings of systems components, graphical modelling, interactions, and implementation considerations, as taught in the unit.

As detailed in this document, to model the system dynamics you are required to develop a systems

design proposal that includes:

- key design requirements derived from the scenario
- a systems architecture represented using graphical modelling notation
- textual descriptions of system components, interactions, and selected technologies used to support system operation

In this assignment, you will prepare an IT system design proposal for a business travel management

system called One World Business Flyer. The system supports business travellers who need to plan and

manage a connected itinerary involving flights, hotels, transport, meetings or events, and nearby

businesses or services.

As an overview, your proposal will focus on:

- defining the logical structure of the system architecture using Fundamental Modelling Concepts

(FMC) notation

- o identifying the user roles, system components, and storage components
- o explaining how those components interact internally and with external systems
- o describing the system components in paragraph form

- defining the implementation aspects of the architecture, including APIs, messaging, and cloud

environment choices

- justifying key design choices in paragraph form

Users interact with one main One World Business Flyer system, while the backend coordinates with

multiple external systems behind the scenes.

1. Systems overview

Business travellers need to attend multi-city business events (e.g., conferences, business partner visits).

For this, they have to coordinate flights, hotels, ground transportation, appointments or events, and nearby

businesses or services relevant to the trip. As part of this, travellers may visit places of interest like tourist

sites, entertainment events, and sports matches.

Decision-making and actions, such as searching, comparing, and booking, are usually spread across

multiple digital platforms. A traveller may search for flights in one platform, hotels in another, maps in

another, and transport options somewhere else. This makes planning fragmented and time-consuming,

especially when timing, proximity, route choices, and nearby options all affect whether the itinerary is valid.

This assignment focuses on the design of an Integrated Business Travel System that supports these

connected travel needs through one coordinated system.

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1.1 About the platform

A fictional travel technology company, One World Business Flyer, wants to develop a system that helps

business travellers plan and manage a connected itinerary.

The system should help users coordinate:

- flights
- hotels
- transportation
- meetings, appointments, or business events
- nearby businesses or services
- points of interest relevant to the trip

These are connected parts of one itinerary. The system is not centred on one service only. It should help

users combine them into a valid trip plan.

1.2 Planning constraints

One World Business Flyer has identified several common problems in current business travel planning:

- relevant information is spread across multiple systems
- users must repeatedly switch between services
- itinerary decisions are connected, but many tools treat them separately
- useful options depend on constraints such as time, distance, route, and proximity
- the traveller needs one coherent itinerary, but the data comes from many places

A valid itinerary may depend on constraints such as:

- arrival or departure time
- distance between locations
- hotel radius from a venue or business site
- transport convenience between key points
- proximity to nearby businesses or services

For example, a traveller may need to choose flights that fit a meeting time, find a hotel within a chosen

radius of the venue, arrange transport between the airport, hotel, and venue, and identify useful nearby

services.

1.3. System backdrop

One World Business Flyer uses a facade pattern as its system backdrop. The user interacts with one main

system, while the backend coordinates with multiple external systems behind the scenes.

These external systems may include:

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- flight information systems
- hotel information systems
- transportation systems
- event or appointment-related systems
- map and location services such as Google Maps
- nearby business or points-of-interest services

You may choose sensible examples of external providers in your proposal. For example, flight information

might come directly from airline systems or from an intermediary such as a Global Distribution System

(GDS). Whether One World Business Flyer connects to a GDS, to individual airline systems, or to both is a

design decision you must make and justify.

The backend should combine information from these sources to support search, recommendation, itinerary

construction, and booking-related decisions.

2. Your task

Prepare a design proposal for the One World Business Flyer system. Your proposal should explain how the

system would support integrated business travel and how its backend would coordinate with connected

external services.

This assessment includes individually led components and a group-integrated component.

The individually led components are Sections 2.1, 2.2, 2.3, 2.5, 2.6, 2.7, and 2.8. Each group must

discuss and agree on a fair distribution of these sections among group members. This distribution must be

listed at the beginning of the submitted PDF in a task allocation table. Each group member must lead one

or more allocated section(s), as identified in that table.

Although these sections are individually led, they must not be developed in isolation.

Group members are

expected to discuss their ideas, share assumptions, and ensure that the components, information objects,

interactions, APIs, external systems, and scenario walkthrough remain consistent across the proposal.

The group-integrated component is Section 2.4 FMC diagram. This section must be completed

collaboratively by the group. The FMC diagram should bring together the individual work from the other

sections into one coherent architecture model. Therefore, the group should finalise the FMC diagram after

the individually led sections have been discussed and aligned.

Each group member is expected to lead their allocated section(s) while also contributing to the overall

consistency of the proposal. The final submission must read as one connected systems design

proposal, not as separate unrelated parts. Use the section headings below to organise your response.

Read the scenario carefully before you begin. Your design should reflect the traveller goals, constraints,

and system needs described in Sections 1.1 to 1.3.

You do not need to list the requirements separately in a formal requirements table.

However, your design

choices throughout the proposal must clearly reflect what the system needs to do for its users and why

those choices are appropriate.

Write your reasoning in paragraph form throughout. You may make reasonable assumptions where

needed, but all assumptions must be listed together at the end of your proposal.

2.1 Scenario interpretation and system goal

Begin by reconstructing the scenario in your own words.

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Write a detailed opening section that:

- explains how you interpret the One World Business Flyer problem
- identifies the main traveller goal
- identifies the main constraints that shape a valid itinerary
- explains what the overall system must achieve at a high level
- states what is inside the One World Business Flyer system boundary and what is outside it as

external systems

- justify each choice logically

This section should show your understanding of the scenario before you describe the architecture.

2.2 Systems and components

Identify the major systems involved, including:

- internal components of the One World Business Flyer system
- external systems it depends on

Your design must include at least:

- one source of flight information
- one source of hotel information
- one source of maps or points-of-interest data

For each internal component you identify, write one separate paragraph that covers:

- the name of the component
- its role in the platform

- the responsibility it handles
- how it connects to other components or external systems where relevant
- why it is needed in your design
- how this component supports one or more of the business traveller's goals described in the scenario

If you discuss external systems individually, explain their role clearly as well.

Your proposal should describe the whole system, not just isolated parts.

2.3 Key information objects

Identify the main information objects the system must work with. Describe them conceptually, not as a

database schema or data structure.

Examples may include:

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- traveller goal
- itinerary
- booking details
- location information
- provider search results
- user details relevant to the trip

For each important information object, briefly explain:

- what it represents
- which parts of the system use it
- why it matters in itinerary planning or coordination

This section is about key information used by the system, not technical data modelling.

2.4 FMC diagram

Create one FMC block diagram that shows:

- the main internal backend components
- the main external systems and services
- the main storage elements used by the platform

- the directional connections between them

Use correct FMC notation as introduced in the unit.

Your diagram must be consistent with the components, systems, and information objects described earlier.

Following your diagram, write one paragraph that traces the path of a typical request through the system,

starting from the point where the traveller enters a goal and ending at the point where a result is returned.

In that paragraph, identify where the facade layer sits, what it does, and which component or interface

represents the point of interaction between the user and the backend.

In the diagram, add a brief label to each internal component indicating whether it is hosted in a public cloud,

private cloud, or on-premise environment. Then write two to three sentences explaining whether the internal

components share the same environment or are distributed across different environments, and why.

2.5 Scenario walkthrough

Provide one end-to-end walkthrough of how the system works for a realistic business-travel situation.

For example, this might begin with a traveller entering a destination and event constraint, continue through

search and coordination with external services, and end with a recommended itinerary or booking-related

result.

This walkthrough should make clear:

- what the traveller is trying to achieve

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- which components are involved
- which external systems are contacted
- what information is passed between parts of the system

- how the platform returns a useful result

This section should explain how the system works, not just what parts exist.

2.6 Interactions and messaging

Identify three to five distinct interactions in the One World Business Flyer backend.

Each interaction must

involve a different pair of components or systems, and you must explain where in the overall system flow

that interaction is used.

If you include messaging infrastructure (queue, topic, event bus), represent it explicitly in FMC and justify

why it is needed. Do not include messaging infrastructure by default.

For each interaction, write one paragraph that:

- names the communicating components or systems
- identifies where that interaction occurs in the system workflow
- classifies the interaction as synchronous, asynchronous, publish-subscribe, or queue-based
- explains why that classification is appropriate given the timing requirements and the nature of the

information being exchanged in that specific communication within the One World Business Flyer

system

The classification alone is not sufficient. Your reasoning for each choice is what demonstrates

understanding of how the system operates.

2.7 API endpoints

Propose six to eight high-level API endpoints that support important system capabilities.

Your endpoints must cover at least:

- one travel search operation
- one location-based retrieval

- one booking-related action
- one itinerary data retrieval

For each endpoint, provide the following as a structured entry:

- HTTP method and endpoint path
- a one-sentence description of what the endpoint does
- key request parameters or request body fields
- key response fields
- the internal component that handles the request

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2.8 Information shared with external systems

Identify at least five types of information that One World Business Flyer must pass to external systems

during the process of searching for options or confirming a booking. For each, write two to three sentences

explaining what the information is, which external system receives it, and why that system needs it in order

to return results or complete the action. Your response should demonstrate that you understand the flow of

information outward from One World Business Flyer to its external providers, not just the flow inward